

CSE-2422 Computer Algorithms Lab

Mapping of Open-ended Problems to the attributes of Complex Engineering Problem

Course code and title	Name of the Problem	CO	PO	WP/EA	Justification
CSE-2422 Computer Algorithms Lab	Students are expected to solve open-ended competitive programming problems where their proficiency in applying algorithms and algorithm design techniques in real life scenario is tested. Their competency is tested both through assignments and timed contest.	CO2	PO2	WP1 WP3 WP2	WP1 (Depth of knowledge required): The given problems cannot be resolved without in-depth engineering fundamentals (K3), engineering specialization (K4) knowledge and requires fundamentals-based, first principles analytical approach. Merely applying an algorithm is insufficient for solving the given problem in most cases. Sufficient customization of the algorithms is needed which is impossible to do without good theoretical understanding and practical experience. WP3 (Depth of analysis required): The given problems have no obvious solution and require abstract thinking, originality in analysis to formulate suitable solution. WP2 (Range of conflicting requirements): Students often require to take decisions in resolving real-world technical constraints while solving problems and analyze performance trade-offs.

Topic: Searching		
#	Problem link	Difficulty
1.	https://codeforces.com/group/MWSDmqGsZm/contest/219774/problem/B	Easy
2.	https://codeforces.com/problemset/problem/1462/A	Easy
3.	https://cses.fi/problemset/task/1621	Easy

Topic: Searching		
#	Problem link	Difficulty
4.	https://codeforces.com/problemset/problem/1744/C	Easy
5.	https://codeforces.com/problemset/problem/1809/B	Easy
6.	https://leetcode.com/problems/kth-largest-element-in-an-array/submissions/1352371907/	Medium
7.	https://codeforces.com/problemset/problem/492/B	Medium
8.	https://codeforces.com/contest/371/problem/C	Medium
9.	https://codeforces.com/problemset/problem/1343/C	Medium
10.	https://cses.fi/problemset/task/1073	Medium
11.	https://cses.fi/problemset/task/1644	Hard
12.	https://cses.fi/problemset/task/1076	Hard
13.	https://codeforces.com/problemset/problem/782/B	Hard
14.	https://cses.fi/problemset/task/2423	Hard
15.	https://cses.fi/problemset/task/1641/	Hard

Topic: Application of some linear and nonlinear DSA		
#	Problem link	Difficulty
1.	https://codeforces.com/group/MWSDmqGsZm/contest/219856/problem/Q	Easy
2.	https://atcoder.jp/contests/arc087/tasks/arc087_a	Easy
3.	https://codeforces.com/contest/1703/problem/D	Easy
4.	https://lightoj.com/problem/double-ended-queue	Easy
5.	https://codeforces.com/contest/1800/problem/C1	Easy
6.	https://codeforces.com/problemset/problem/26/B	Medium
7.	https://leetcode.com/problems/longest-increasing-subsequence/description/	Medium
8.	https://leetcode.com/problems/longest-increasing-subsequence/description/	Medium
9.	https://codeforces.com/problemset/problem/676/C	Medium
10.	https://cses.fi/problemset/task/1076/	Medium
11.	https://www.spoj.com/problems/ORDERSET/en/	Hard
12.	https://cses.fi/problemset/task/2428	Hard
13.	https://leetcode.com/problems/sliding-window-maximum/description/	Hard

Topic: Sorting		
#	Problem link	Difficulty
1.	https://leetcode.com/problems/merge-sorted-array/description/	Easy
2.	https://leetcode.com/problems/maximum-gap/description/	Easy
3.	https://leetcode.com/problems/3sum/description/	Easy
4.	https://leetcode.com/problems/largest-divisible-subset/description/	Easy
5.	https://codeforces.com/problemset/problem/632/C	Easy
6.	https://codeforces.com/problemset/problem/1986/C	Easy

Topic: Dynamic programming (Matrix chain multiplication, LCS, LIS, and other Basic DP)			
#	Problem link	Tag	Difficulty
1.	https://atcoder.jp/contests/tessoku-book/tasks/dp_a?lang=en		Easy
2.	https://leetcode.com/problems/coin-change-ii/description/	Coin change	Easy
3.	https://leetcode.com/problems/coin-change/description/	Coin change	Easy
4.	https://cses.fi/problemset/task/1637	0/1 knapsack	Easy
5.	https://atcoder.jp/contests/dp/tasks/dp_d	DP on grid	Easy
6.	https://atcoder.jp/contests/dp/tasks/dp_c?lang=en		Easy
7.	https://lightoj.com/problem/coin-change-i	Coin change	Medium
8.	https://atcoder.jp/contests/dp/tasks/dp_e	Knapsack	Medium
9.	https://cses.fi/problemset/task/1639/		Medium
10.	https://atcoder.jp/contests/dp/tasks/dp_f?lang=en	LCS	Medium

Topic: Basic Greedy		
#	Problem link	Difficulty
1.	https://www.spoj.com/problems/BUSYMAN/	Easy
2.	https://cses.fi/problemset/task/1632	Easy
3.	https://cses.fi/problemset/task/1140	Easy
4.	https://vjudge.net/problem/Aizu-ALDS1_15_B	Easy
5.	https://vjudge.net/problem/UVA-1072	Easy
6.	https://codeforces.com/contest/79/problem/A	Easy
7.	https://codeforces.com/gym/101149/problem/B	Easy
8.	https://open.kattis.com/problems/piecesofparentheses	Easy
9.	https://atcoder.jp/contests/cf17-final/tasks/cf17_final_d	Easy
10.	https://codeforces.com/contest/1013/problem/B	Easy
11.	https://codeforces.com/contest/58/problem/B	Easy
12.	https://codeforces.com/contest/1009/submission/199871754	Easy
13.	https://codeforces.com/contest/853/problem/A	Easy
14.	https://codeforces.com/contest/527/problem/B	Easy
15.	https://codeforces.com/contest/45/problem/I	Easy

Topic: Elementary graph (BFS, DFS)			
#	Problem link	Topic	Difficulty
1.	https://cses.fi/problemset/task/1666	component count	Easy

Topic: Elementary graph (BFS, DFS)			
#	Problem link	Topic	Difficulty
2.	https://cses.fi/problemset/task/1674	subtree vertex count	Easy
3.	https://codeforces.com/problemset/problem/580/C	subtree	Easy
4.	https://cses.fi/problemset/task/1667	shortest path bfs	Easy
5.	https://www.spoj.com/problems/NAKANJ/	2d grid visit	Easy
6.	https://codeforces.com/contest/24/problem/A	basic dfs	Easy
7.	https://codeforces.com/contest/802/problem/J	dfs tree	Easy
8.	https://cses.fi/problemset/task/1679	topological sort	Easy
9.	https://cses.fi/problemset/task/1668	bipartite graph	Easy
10.	https://www.spoj.com/problems/BUGLIFE/	bipartite graph	Easy
11.	https://codeforces.com/contest/862/problem/B	bipartite graph	Easy
12.	https://codeforces.com/contest/378/problem/C	dfs on 2d grid	Medium
13.	https://codeforces.com/contest/769/problem/C	dfs on 2d grid	Medium
14.	https://cses.fi/problemset/task/1192	component count	Hard
15.	https://atcoder.jp/contests/abc291/tasks/abc291_e	topological sort	Hard

Topic: MST (Prim, Kruskal)		
#	Problem link	Difficulty
1.	https://www.spoj.com/problems/KOICOST/	Easy
2.	https://www.spoj.com/problems/MARYBMW/	Easy
3.	https://www.hackerrank.com/contests/june-world-codesprint/challenges/johnland/problem	Easy
4.	https://www.codechef.com/problems/CHEFELEC	Easy
5.	https://usaco.org/index.php?page=viewproblem2&cpid=531	Easy

Topic: Single-source shortest paths (Dijkstra, Bellman-Ford)		
#	Problem link	Difficulty
1.	http://codeforces.com/problemset/problem/20/C	Easy
2.	https://vjudge.net/problem/SPOJ-SHPATH	Easy
3.	https://lightoj.com/problem/country-roads	Easy
4.	https://cses.fi/problemset/task/1671	Easy
5.	https://cses.fi/problemset/task/1673	Easy
6.	https://cses.fi/problemset/task/1195	Easy

Topic: Single-source shortest paths (Dijkstra, Bellman-Ford)		
#	Problem link	Difficulty
7.	https://atcoder.jp/contests/abc361/tasks/abc361_d	Medium
8.	https://cses.fi/problemset/task/1197/	Medium
9.	https://codeforces.com/contest/1846/problem/G	Hard

Topic: All-pairs shortest paths (Floyd-Warshall)		
#	Problem link	Difficulty
1.	https://www.spoj.com/problems/ROHAAN/en/	Easy
2.	https://cses.fi/problemset/task/1672	Easy
3.	https://codeforces.com/problemset/problem/295/B	Medium
4.	https://toph.co/p/kingdom-of-code	Hard
5.	https://codeforces.com/contest/25/problem/C	Hard
6.	https://www.spoj.com/problems/ARBITRAG/	Hard
7.	https://codeforces.com/gym/101845/problem/C	Hard
8.	https://codeforces.com/contest/33/problem/B	Hard

Topic: Backtracking (N-Queens and other basic backtracking)		
#	Problem link	Difficulty
1.	https://leetcode.com/problems/binary-tree-paths/description/	Easy
2.	https://www.hackerearth.com/practice/basic-programming/recursion/recursion-and-backtracking/practice-problems/algorithm/n-queensrecursion-tutorial/	Easy
3.	https://leetcode.com/problems/n-queens/description/	Easy
4.	https://cses.fi/problemset/task/1622	Easy
5.	https://leetcode.com/problems/binary-watch/description/	Medium
6.	https://www.codechef.com/problems/J3	Medium
7.	https://www.geeksforgeeks.org/problems/m-coloring-problem-1587115620/1	Medium
8.	https://www.geeksforgeeks.org/problems/rat-in-a-maze-problem/1	Medium
9.	https://www.hackerearth.com/practice/basic-programming/recursion/recursion-and-backtracking/practice-problems/algorithm/nearby-squares-338a4a64/	Medium
10.	https://www.codechef.com/problems/J1	Medium
11.	https://leetcode.com/problems/sudoku-solver/description/	Hard
12.	https://codeforces.com/problemset/problem/476/B	Hard
13.	https://cses.fi/problemset/task/2165	Hard
14.	https://onlinejudge.org/index.php?option=onlinejudge&page=show_problem&problem=569	Hard
15.	https://onlinejudge.org/index.php?option=com_onlinejudge&Itemid=8&category=24&page=show_problem&problem=691	Hard

Topic: Computational geometry (line segment properties, line segment intersection, convex hull with Graham's scan)		
#	Problem link	Difficulty
1.	https://codeforces.com/group/MWSDmqGsZm/contest/223338/problem/T	Easy
2.	https://codeforces.com/group/MWSDmqGsZm/contest/223338/problem/U	Easy
3.	https://codeforces.com/group/MWSDmqGsZm/contest/223338/problem/V	Easy
4.	https://codeforces.com/group/MWSDmqGsZm/contest/223338/problem/X	Easy
5.	https://codeforces.com/group/MWSDmqGsZm/contest/223338/problem/Z	Easy
6.	https://codeforces.com/group/MWSDmqGsZm/contest/223338/problem/W	Medium
7.	https://open.kattis.com/problems/convexhull	Medium
8.	https://open.kattis.com/problems/parade	Medium
9.	https://codeforces.com/group/MWSDmqGsZm/contest/223338/problem/Y	Medium

Topic: Number theory (GCD, Modular exponentiation, RSA, and other basic number theory problem)		
#	Problem link	Difficulty
1.	https://www.codechef.com/problems/PRB01	Easy
2.	https://codeforces.com/group/MWSDmqGsZm/contest/223338/problem/H	Easy
3.	https://codeforces.com/group/MWSDmqGsZm/contest/223338/problem/Q	Easy
4.	https://codeforces.com/contest/1985/problem/B	Easy
5.	https://codeforces.com/contest/1999/problem/F	Easy
6.	https://codeforces.com/group/MWSDmqGsZm/contest/223338/problem/J	Medium
7.	https://codeforces.com/problemset/problem/742/A	Medium
8.	https://www.codechef.com/problems/GCDMOD	Medium
9.	https://codeforces.com/problemset/problem/1968/A	Medium
10.	https://codeforces.com/contest/26/problem/A	Medium
11.	https://onlinejudge.org/index.php?option=com_onlinejudge&Itemid=8&category=24&page=show_problem&problem=484	Hard
12.	https://www.spoj.com/problems/TDKPRIME/	Hard
13.	https://www.spoj.com/problems/PRIME1/	Hard
14.	https://codeforces.com/problemset/problem/1954/C	Hard
15.	https://codeforces.com/contest/1780/problem/B	Hard
16.	https://codeforces.com/contest/1360/problem/D	

Topic: String matching algorithms			
#	Problem link	Tag	Difficulty
	https://codeforces.com/edu/course/2/lesson/2	Suffix Array practice problem from step 1-3	Easy
1.	https://www.spoj.com/problems/SUFEQPRE/	Z function	Easy
2.	https://cses.fi/problemset/task/1753	Hashing	Easy
3.	https://cses.fi/problemset/result/9240719/	Hashing	Easy
4.	https://www.spoj.com/problems/NAJPF/	KMP	Easy
5.	https://codeforces.com/contest/271/problem/D	Hashing	Medium
6.	https://codeforces.com/problemset/problem/126/B	Hashing	Medium
7.	https://codeforces.com/edu/course/2/lesson/2	Suffix Array practice problem from step 4, first 3 problem from step 5 can be tried also	Medium
8.	https://cses.fi/problemset/task/1733	Hashing	Medium
9.	https://lightoj.com/problem/substring-frequency	SUFFIX ARRAY	Medium
10.	https://codeforces.com/contest/1968/problem/G1	Hashing	Hard
11.	https://codeforces.com/contest/727/problem/E	Hashing	Hard
12.	https://codeforces.com/problemset/problem/149/E	Z function	Hard
13.	https://cses.fi/problemset/task/1110/	Hashing	Hard
14.	https://cses.fi/problemset/task/1111	Hashing	Hard